

Participant-Level PHQ-8 Aggregation: Mathematical Formulation

Step 1: Utterance Segmentation

$$A_p = \{A_1, A_2, \dots, A_n\} \quad \text{where } n = \text{number of utterances for participant } P$$

Step 2: Per-Utterance PHQ-8 Prediction (Regression)

$$PHQ8_hat_i = f_regression(\text{SpeechEncoder}(A_i) , \text{FusionLayer}, \text{Classifier})$$

Step 3: Mean Aggregation over Utterances

$$PHQ8_hat_P = (1/n) * \sum(PHQ8_hat_i) \quad \text{for } i \text{ in } \{1, \dots, n\}$$

Step 4: Binary Classification from Predicted Score

$$\text{Label}_P = 1 \text{ (Depressed)} \quad \text{if } PHQ8_hat_P \geq \theta, \quad \text{else } 0 \quad [\theta = 10]$$

Regression Loss (Training)

$$L_MSE = (1/N_P) * \sum((PHQ8_hat_P - PHQ8_true_P)^2) \quad \text{over all participants } P$$

Evaluation Metrics

$$MAE=2.537 \quad RMSE=3.348 \quad \text{Pearson_r}=0.858 \quad CCC=0.857 \quad R^2=0.704$$